



Relationship between the characteristic temperature and terahertz permittivity of different rank coals considering FeS₂ content

Baolin Qu^a, Jingxin Wang^a, Hongqing Zhu^{a,*}, Qi Liao^a, Lintao Hu^a, Haoran Wang^b

^a School of Emergency Management and Safety Engineering, China University of Mining and Technology-Beijing, Beijing 100083, PR China

^b Department of Automation, Tsinghua University, Beijing 100084, PR China

ARTICLE INFO

Keywords:

Lignite
Pyrite
Characteristic temperature
Permittivity
Terahertz
Mathematical relationship

ABSTRACT

In this study, the characteristic temperatures and terahertz permittivity of lignite with varying pyrite content were determined using a synchronous thermal analyzer and terahertz time-domain spectroscopy. A mathematical relationship between these two parameters was established by combining experimental data. The results indicate that within the range of 0–9% pyrite content, there is a trend where pyrite initially enhances and then weakens the promotion effect on coal spontaneous combustion. A quadratic polynomial can describe this relationship, with the extremum occurring at 6%. Regarding permittivity, as the pyrite content increases from 0% to 9%, the real part of permittivity exhibits irregular variations, while the imaginary part shows a well-defined linear increase. This behavior is attributed to enhanced relaxation polarization, leading to more significant dielectric losses. Furthermore, through the combination of experimental data, a robust quadratic relationship is observed between characteristic temperature and terahertz permittivity. The credibility of this relationship is validated by comparing it with experimental results. These findings contribute to the advancement of the theoretical understanding of coal's permittivity and spontaneous combustion characteristics, and provide a theoretical possibility for the rapid estimation of spontaneous combustion characteristics in pyrite-bearing coal seams using electromagnetic methods.

1. Introduction

In recent decades, coal spontaneous combustion (CSC) has emerged as a substantial risk within global coal mining operations [1,2]. This hazard not only leads to the depletion of valuable resources but also results in the release of significant quantities of toxic gases, leading to environmental pollution and posing threats to human safety [3,4]. Beyond external environmental factors, the likelihood of CSC is influenced by various internal factors, including moisture content, coal rank, pore structure, and mineral composition [5–7]. Among these factors, the presence of pyrite (FeS₂) is widely acknowledged as a primary catalyst for CSC [8–10]. As the demand for coal continues to rise, expanded mining activities in terms of both area and depth have led to a notable increase in the extraction of coal containing substantial pyrite content [11]. Consequently, it is imperative to conduct precise assessments and implement effective controls to manage the risks associated with the spontaneous combustion of pyrite-rich coal.

In recent years, numerous scholars have dedicated their research efforts to understanding the influence and underlying mechanisms of

pyrite content on CSC [12–14]. However, there has been a noticeable oversight in the development of rapid and precise methods for detecting pyrite content within coal seams. It is established that pyrite exists in various forms, primarily as fine-grained particles with uneven distribution throughout coal seams [15,16]. Consequently, this non-uniform distribution leads to significant variations in pyrite content among coals within the same seam [17]. This diversity in pyrite content translates to notable differences in the spontaneous combustion characteristics of coals within the same seam. For coal seams rich in pyrite, the current approaches to coal fire prevention and control, such as gelling, nitrogen injection, and grouting, however, tend to generalize the properties of these coals based on their overall tendencies towards spontaneous combustion [18–20]. This generalized approach can result in localized CSC risks and unnecessary consumption of firefighting resources. Hence, the efficient estimation of pyrite content distribution within coal seams holds immense importance for precise CSC risk management.

The electromagnetic method, which relies on disparities in electromagnetic properties between the target medium and its surroundings,

* Corresponding author.

E-mail address: zhq@cumtb.edu.cn (H. Zhu).

<https://doi.org/10.1016/j.infrared.2023.104995>

Received 7 August 2023; Received in revised form 27 September 2023; Accepted 11 November 2023

Available online 13 November 2023

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offers several advantages, including high efficiency, cost-effectiveness, wide applicability, and non-destructive capabilities [21]. This method has found extensive utility in various fields, including geological exploration, coal fire detection, and environmental monitoring [22–24]. Central to this method is permittivity, a crucial physical parameter that characterizes the electromagnetic properties of a medium. It reflects the medium's ability to respond to electromagnetic waves through dielectric or polarizing properties. Notably, there are discernible distinctions in the dielectric characteristics between pyrite and coal, making them potentially distinguishable through electromagnetic methods. Furthermore, permittivity is directly linked to the frequency of electromagnetic waves [25]. In recent years, there has been a surge of interest in terahertz (THz) waves, which encompass electromagnetic waves with radiation frequencies ranging from 0.1 to 10 THz. These waves overlap with both the millimeter wave in long-wavelength regions and the infrared wave in short-wavelength regions [26]. Terahertz waves offer unique advantages, including coherence, broad bandwidth, and high transient transmittance, making them invaluable for imaging and spectral identification across various domains. Terahertz waves, in particular, exhibit sensitivity to diverse intermolecular, lattice, and skeletal vibrations, with a pronounced focus on organic materials. This sensitivity enables the provision of structural information and the determination of material composition. Numerous studies have already demonstrated the practicality of terahertz technology in the realms of energy and fuel research [27–31], thereby opening novel avenues for material identification. It's essential to note that traditional detection methods, such as X-ray diffraction, suffer from drawbacks like lengthy scanning times and challenges in quantitative analysis [32]. Consequently, this study delves into the influence of pyrite on the permittivity of coal within the terahertz range. This investigation holds the potential to provide valuable theoretical support for discerning the distribution of pyrite content within coal seams, thus enhancing our ability to manage and mitigate combustion risks associated with pyrite-bearing coal.

Moreover, it is essential to indirectly assess coal's propensity for spontaneous combustion through its terahertz permittivity. To date, common laboratory techniques for directly measuring coal's spontaneous combustion characteristics encompass methods such as the oxygen adsorption method, crossover point temperature method, basket calorimetry method, adiabatic oxidation method, and thermogravimetric analysis method. However, despite their relative accuracy, these methods suffer from certain limitations, including lengthy testing durations and the necessity for a substantial number of samples. In the coal seams containing pyrite, the pyrite content plays a significant role in shaping both their spontaneous combustion characteristics and permittivity [12,14,33]. Hence, it is reasonable to expect a numerical relationship between these two variables. Consequently, our study employed synchronous thermal analyzers and terahertz time-domain spectroscopy to determine the characteristic temperature and permittivity of lignite samples with varying pyrite content. Subsequently, we established a mathematical relationship between these parameters by analyzing experimental data. By using pyrite content as a bridge, our analysis revealed that the relationship between characteristic temperatures and permittivity can be effectively described by a quadratic polynomial function. Finally, we validated the reliability of this relationship by comparing it with experimental results. These findings not only contribute to the advancement of theoretical knowledge regarding permittivity and spontaneous combustion characteristics in coal but also offer a theoretical foundation for the rapid estimation of spontaneous combustion tendencies in pyrite-bearing coal seams using electromagnetic methods.

2. Experimental methods

2.1. Coals

The experimental coals include lignite, long flame coal, non-caking

coal, and 1/3 coking coal. Coal samples were prepared in accordance with the Chinese National Standard, GB474-2008. Following preparation, proximate analysis was conducted based on the Chinese National Standard, GB/T 212-2008, and the results are presented in Table 1. The oxide layer on the coal surface was removed, and the coal samples were subsequently crushed to obtain pulverized coal with a particle size of less than 74 μm . Pyrite is often found embedded in coal in the form of fine-grained particles [16]; thus, FeS_2 powder and pulverized coal were blended in specific mass ratios to make coal samples with varying mass percentages of pyrite (0 %, 1 %, 2 %, 3 %, 6 %, and 9 %). The mass percentages of pyrite were determined according to the Chinese national standard, GB/T 15224.2-2021. The FeS_2 powder used in the experiment boasts a purity of 99.9 % and a particle size of less than 74 μm . All prepared samples were vacuum-sealed for storage.

2.2. TG analysis

The characteristic temperatures of coal samples were determined using a synchronous thermal analyzer (STA 449F3, NETZSCH) with an accuracy of 10^{-4} mg. Each experiment utilized precisely 10 ± 0.1 mg of coal samples. To simulate an air atmosphere, a mixture of O_2 (10 ml/min) and N_2 (40 ml/min) was introduced simultaneously in a ratio of 1:4. The samples were heated from 25 $^\circ\text{C}$ to 850 $^\circ\text{C}$ at a heating rate of 5 K/min.

2.3. THz-TDS analysis

(1) Tableting

Due to the limited penetration capability of terahertz waves, high-density polyethylene (HDPE) was employed to dilute coal samples with varying pyrite content at a mass ratio of 2:1 [29]. The total mass of the mixture was approximately 0.27 g. The HDPE used in this study had a particle size of less than 30 μm and exhibited high transparency in the terahertz band [34]. The resulting mixture was subjected to a pressure of 10 MPa for 2 mins, resulting in a compressed sample in the form of a tablet with a diameter of approximately 13 mm and a thickness of about 2 mm.

(2) THz-TDS system

The permittivity of coal samples in the terahertz band was tested by a terahertz time-domain spectrometer (FICO THz-TDS system, Zomega Terahertz Corporation). The system can measure far-infrared spectra from 0.1 to 3.0 THz with a peak dynamic range of greater than 1000 (60 dB) [35]. The experiments were conducted within a closed workbench under controlled conditions, maintaining an ambient temperature of 25 $^\circ\text{C}$ and humidity below 3 % [28]. The samples were placed in arrays on a test platform for transmissive spectral acquisition. The size of the spot is 2 mm.

(3) THz data processing

Before data acquisition, the FICO THz-TDS system is powered on for 30 min to ensure signal and environmental stability. The THz-TDS system directly captures time-domain signals of transmitted terahertz waves. Dry air data serves as the reference signal, while the data of terahertz waves passing through the sample is used as the sample signal. The time-domain signal obtained is subsequently converted into a frequency-domain signal using a fast Fourier transform (FFT). Based on the phase and amplitude information of the signal, critical optical and electromagnetic parameters of the sample, including refractive index n ,

Table 1
Proximate analysis of coals.

Coal	Fixed carbon/%	Volatile/%	Moisture/%	Ash/%
HM	56.63	22.93	12.82	7.62
CYM	34.26	20.73	12.52	32.49
BNM	50.99	24.81	4.20	20.00
1/3JM	52.04	32.96	1.35	13.65

extinction coefficient K , real permittivity ϵ' , and imaginary permittivity ϵ'' , can be calculated using Eqs. (1)–(4), with detailed data processing methods outlined in references [36,37].

$$n = \frac{\Delta\varphi c}{\omega d} + 1 \quad (1)$$

$$\kappa = \ln \left[\frac{4n}{p \cdot (n+1)^2} \right] \frac{c}{\omega d} \quad (2)$$

$$\epsilon' = n^2 - \kappa^2 \quad (3)$$

$$\epsilon'' = 2n\kappa \quad (4)$$

3. Results and discussion

3.1. Characteristic temperature of lignite with different FeS_2 content by TG analysis

Fig. 1 shows the TG (Thermogravimetric) and DTG (Derivative Thermogravimetric) curves of lignite coal during the heating process. Characteristic temperatures play a crucial role in describing the spontaneous combustion behavior of lignite. In accordance with previous studies [28,38,39], seven characteristic temperatures can be observed in Fig. 1. T_1 is the temperature at which the coal exhibits the maximum rate of dehydration and gas release. T_2 is the temperature at which the coal mass begins to increase, signifying the initiation of chemical oxygen absorption. T_3 is the temperature at which the coal displays the maximum rate of mass increase. T_4 is the temperature at which the mass reaches its peak value, marking the point where the rate of change in mass due to adsorbed oxygen and decomposition oxidation products attains dynamic equilibrium. T_5 is the temperature of coal combustion. T_6 is the temperature at which the coal shows the maximum rate of mass loss during the combustion process; T_7 is the temperature at which the coal is completely burned out. Both T_5 and T_7 are determined using the tangent method [40].

Based on these characteristic temperatures, the oxidation process of coal can be divided into five stages. These five stages are distinguished in Fig. 1(a) by different colors. The first stage is the dehydration and gas release stage (T_0 – T_2), during which the coal loses water and releases gases, resulting in a decrease in coal mass. The second stage is the oxygen absorption and weight gain stage (T_2 – T_4), during which functional groups in the coal chemically adsorb oxygen, resulting in an increase in coal mass. The third stage is the oxidative decomposition stage (T_4 – T_5),

during which the rate of oxidation reaction gradually accelerates, releasing organic gases. The fourth stage is the oxidative combustion stage (T_5 – T_7), during which the cyclic reaction rate dramatically increases, and the cyclic structures decompose and burn, releasing heat and generating gaseous products, resulting in a rapid loss of mass. After the fourth stage, the coal is converted to ash and its weight remains relatively stable.

Fig. 2 shows the TG and DTG curves of lignite with varying FeS_2 content. Utilizing the aforementioned methodology, Table 2 presents the characteristic temperatures of lignite with varying FeS_2 content. It is evident that pyrite plays a pivotal role in determining these characteristic temperatures. As FeS_2 content increases, the ash content proportionally increases post-burnout, confirming the uniformity of sample mixing. Within the range of 0 % to 9 % FeS_2 content, pyrite exhibits a pattern of initially enhancing and subsequently diminishing the promotion effect on coal spontaneous combustion, with the peak occurring at 6 %. At this critical concentration, the characteristic temperature T_2 —marking the termination of the dehydration and degassing stage and the initiation of the oxygen absorption and weight gain stage—shifts from 138.38 °C to 106.94 °C. This shift is attributed to the ability of FeS_2 to reduce coal moisture content, oxidize, release heat under the action of moisture, reduce moisture evaporation, and increase moisture temperature in advance [8]. Additionally, the characteristic temperature T_4 —representing the termination of the oxygen absorption and weight gain stage and the initiation of the oxidative decomposition stage—shifts from 281.28 °C to 288.14 °C. Similarly, the characteristic temperature T_5 —indicating the termination of the oxidative decomposition stage and the initiation of the oxidative combustion stage—shifts from 426.39 °C to 432.77 °C. In summary, the promotion effect of FeS_2 on the spontaneous combustion stage of coal manifests as the premature termination of the dehydration and degassing stage, the early initiation and delayed termination of the oxygen absorption and weight gain stage, the delayed initiation and delayed termination of the oxidative decomposition stage, and the delayed initiation and premature termination of the oxidative combustion stage.

Moreover, it can be observed that pyrite makes the oxygen absorption and weight gain stage on the TG curve more obvious. At 6 % FeS_2 , the maximum value of TG reaches from 93.67 % at 0 % FeS_2 to 101.43 %, even exceeding the original mass of coal, indicating that pyrite can enhance the oxygen uptake of coal. According to previous studies, the effects of FeS_2 on coal spontaneous combustion encompass both its exothermic reactions and its effects on the microstructure and catalytic processes within coal. Pyrite can indeed undergo exothermic reactions during coal spontaneous combustion. These exothermic reactions release heat energy, and as they accumulate, they contribute to the overall macroscopic calorific value of the coal [5,9]. Furthermore, beyond its own heat release, pyrite can also affect the microstructure of coal, influencing factors such as pore structure, oxygen uptake, and catalysis. While pyrite may not directly alter the pore structure of raw coal, its heat release can promote the expansion and cracking of the coal matrix, leading to more developed coal pores. This increased pore development, in turn, enhances the oxygen uptake of the coal [8], further resulting in a prolonged reaction stage and a significant increase in coal mass. Additionally, pyrite and its oxidation products exhibit significant catalytic effects on the coal oxidation process, particularly through species such as Fe^{3+} , Fe^{2+} , and H^+ . These elements play roles in accelerating the oxidation of specific coal structures (e.g., $-\text{CH}_2-$ and $-\text{OH}$), leading to the formation of $-\text{COOH}$. Furthermore, H^+ enhances the low-temperature oxidation of coal by increasing the adsorption of oxygen by coal [6].

To obtain quantitative information, Fig. 3 illustrates the relationship between FeS_2 content and characteristic temperatures T_2 , T_4 , and T_5 . These exhibit a clear and well-defined pattern that can be accurately represented by a quadratic polynomial function, $T = a_1 + b_1x + c_1x^2$, as outlined in Table 3. The correlation coefficients (R^2) for these fits are 0.98, 0.99, and 0.98, respectively. These high correlation coefficients

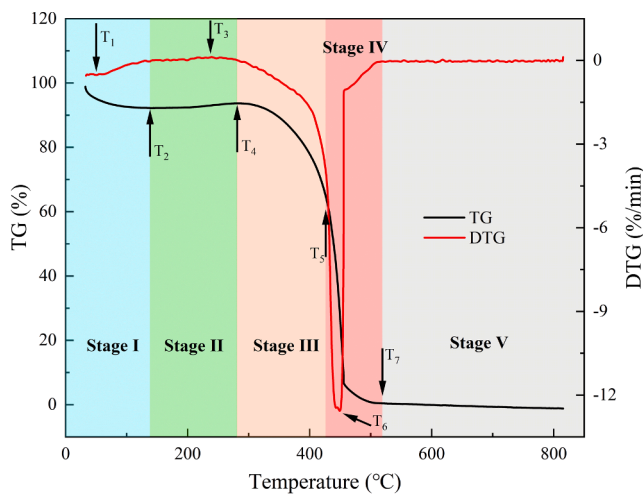


Fig. 1. Seven characteristic temperatures and five stages of coal. Blue, green, orange, red, and gray areas represent the first through fifth stages of coal spontaneous combustion, respectively.

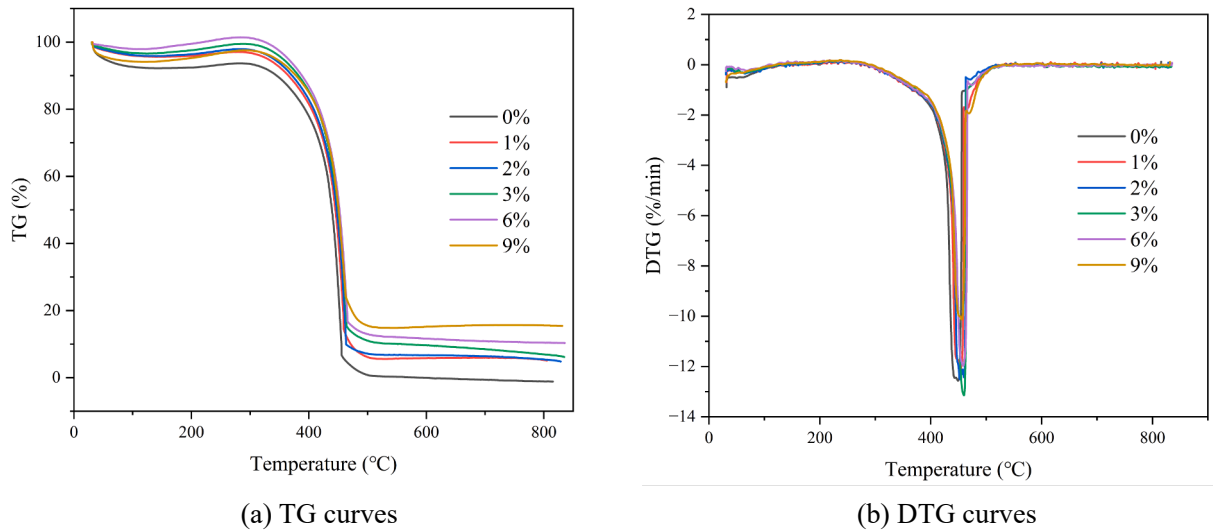


Fig. 2. (a) TG and (b) DTG curves of lignite. Curves were measured using lignite with different pyrite content (0 %, 1 %, 2 %, 3 %, 6 %, and 9 %).

Table 2
Characteristic temperatures of lignite with different FeS₂ content.

FeS ₂ content/%	Characteristic temperature (°C)						
	T ₁	T ₂	T ₃	T ₄	T ₅	T ₆	T ₇
0	60.48	138.38	237.34	281.28	426.39	449.28	519.38
1	63.75	134.45	237.34	283.25	428.29	453.05	518.26
2	65.01	131.82	238.24	285.11	429.09	452.51	517.11
3	64.03	124.63	237.43	286.53	430.72	458.93	515.34
6	68.14	106.94	238.34	288.14	432.77	459.34	512.14
9	60.04	119.64	237.34	287.64	429.81	455.84	518.44

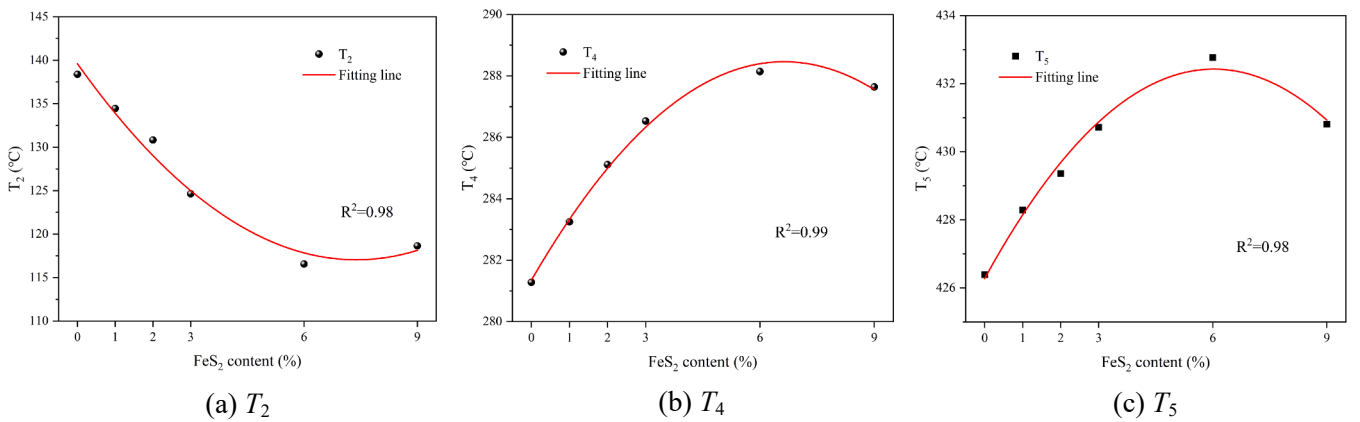


Fig. 3. Variation of characteristic temperature of (a) T₂, (b) T₄, and (c) T₅ versus FeS₂ content. Relationship is fitted by function of quadratic polynomial.

Table 3
Fitting results of characteristic temperature and FeS₂ content.

Characteristic temperature	$T = a_1 + b_1x + c_1x^2$			
	a_1	b_1	c_1	R^2
T ₂	139.59	-6.11	0.41	0.98
T ₄	281.35	2.15	-0.16	0.99
T ₅	426.28	2.04	-0.17	0.98

demonstrate that the characteristic temperature of coal can be reliably estimated based on the FeS₂ content using the empirical formula.

3.2. Terahertz permittivity of lignite with different FeS₂ content by THz-TDS analysis

(1) Determination for the effective frequency band of lignite

According to the previous study [41], it has been established that the dielectric properties, as measured by THz-TDS systems, are subject to the influence of frequency and device accuracy under electromagnetic fields. Generally, higher frequencies offer improved resolution but lower signal strength. Considering that coal is inherently a polar substance, high-frequency electromagnetic waves tend to attenuate as they

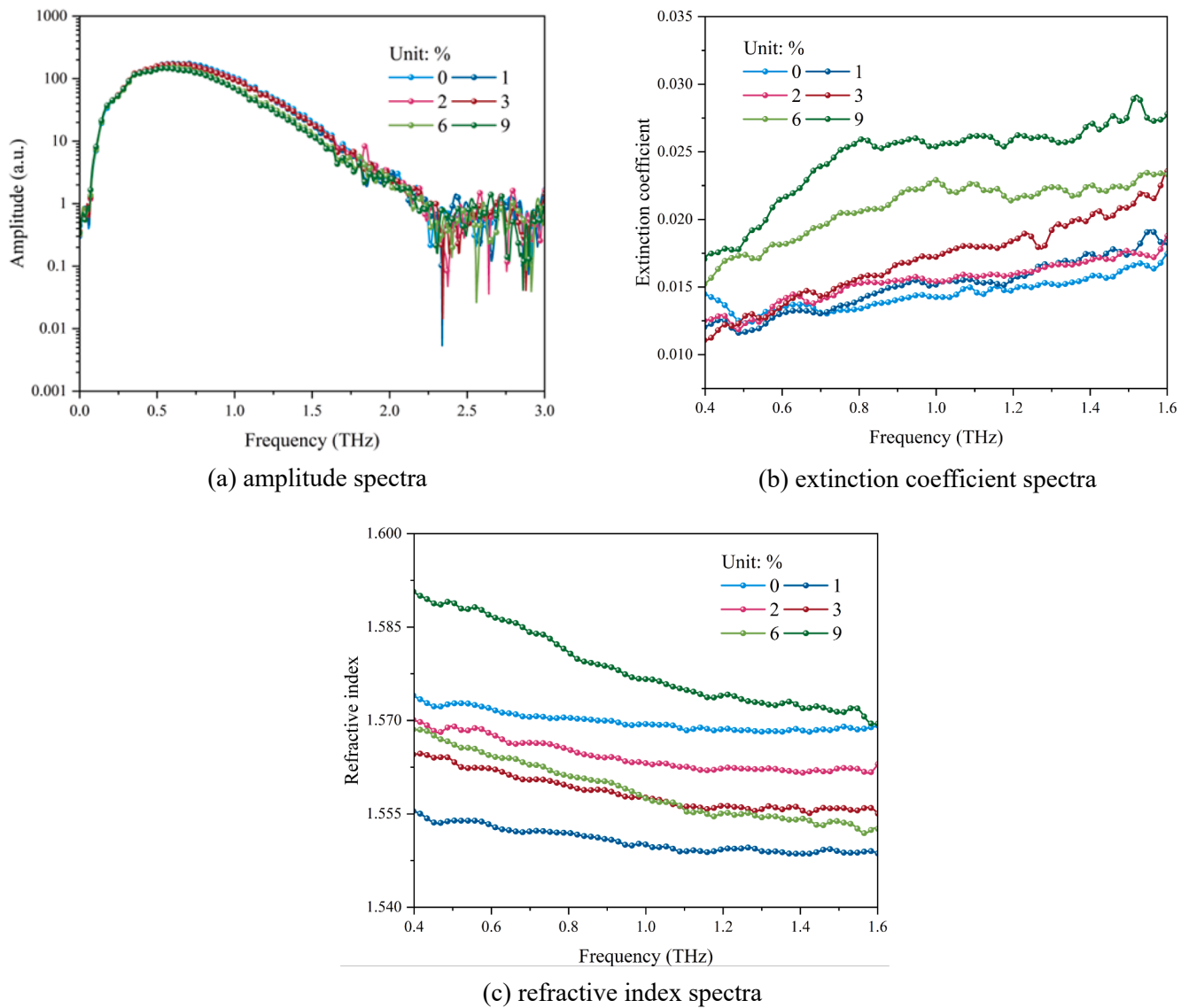


Fig. 4. (a) amplitude spectra, (b) extinction coefficient spectra, and (c) refractive index spectra of lignite with different FeS₂ content. Spectra was measured using lignite with different pyrite content (0 %, 1 %, 2 %, 3 %, 6 %, and 9 %).

penetrate coal samples. Therefore, it becomes essential to select an effective frequency band for analysis. This effective frequency band can be identified through terahertz amplitude spectra. As depicted in Fig. 4, which illustrates terahertz frequency-domain spectra of lignite with varying pyrite content (0 %, 1 %, 2 %, 3 %, 6 %, and 9 %), it is evident that exceptional signal quality is achieved within the 0.4–1.6 THz range, while other bands exhibit substantial background noise levels that hinder accurate analysis. To further illustrate this point, extinction coefficient spectra and refractive index spectra at 0.4–1.6 THz are displayed in Fig. 4(b) and (c). As shown in the figures, the extinction coefficient generally increases with frequency, while the refractive index exhibits an overall declining trend. Notably, within the band range of 0.4–1.7 THz, these curves exhibit clear demarcations and superior signal quality. As a result, the dielectric spectrum of lignite within the 0.5–1.5 THz frequency region was selected for further analysis.

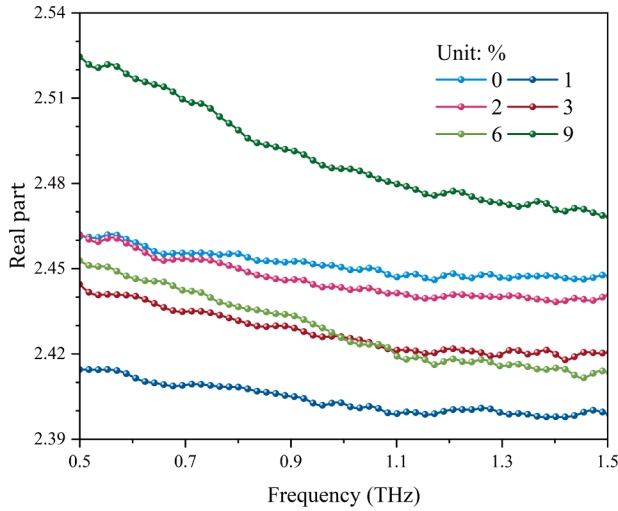
(2) Terahertz permittivity of lignite

Permittivity is a vital physical parameter that characterizes the electromagnetic properties of coal, and its expression is detailed in Eq. (5). The real part of permittivity represents energy storage and is positively correlated with polarity, while its imaginary part signifies energy dissipation, reflecting the absorption of terahertz waves. Owing to the

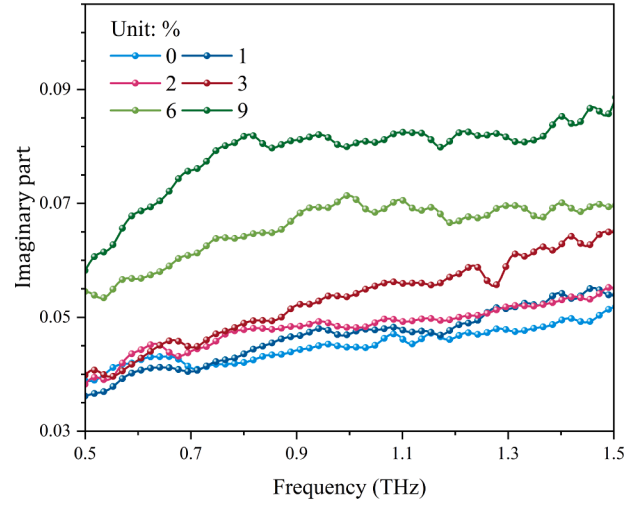
superior signal-to-noise ratio (SNR), the dielectric spectrum of lignite with varying FeS₂ content in the 0.5–1.5 THz frequency range is presented in Fig. 5. Noticeable differences are observed in the dielectric spectrum of coal samples, underscoring the ability of THz-TDS to detect variations in coal's dielectric properties attributed to differing FeS₂ content.

$$\varepsilon = \varepsilon' + j\varepsilon'' \quad (5)$$

Fig. 5(a) displays the real part of the terahertz permittivity of lignite with varying FeS₂ content. The real part exhibits irregular changes as FeS₂ content increases. This irregular behavior can be attributed to the influence of lattice defects and mixed valence states within FeS₂ crystals, which belong to ionic crystals. Under the influence of terahertz fields, these crystals generate both ionic polarization and relaxation polarization. Notably, the intensity of ion polarization is significantly lower than that of orientation polarization, which predominantly governs the polarity of coal [30,42]. Consequently, changes in polarity are relatively weak. Another significant factor is the heterogeneity of coal, which results in variations in pore development and distribution, significantly impacting polarity [43]. In addition, the real part exhibits abnormal dispersion as the frequency changes. This phenomenon can be attributed



(a) Real part



(b) Imaginary part

Fig. 5. (a) real part and (b) imaginary part of terahertz permittivity of lignite. Permittivity was measured using lignite with different pyrite content (0 %, 1 %, 2 %, 3 %, 6 %, and 9 %).

to relaxation polarization, with the dispersion amplitude of the real part gradually increasing with higher FeS_2 content.

Fig. 5(b) illustrates the imaginary part of the terahertz permittivity of lignite with varying FeS_2 content. Generally, coals with higher FeS_2 content exhibit higher imaginary parts owing to more pronounced dielectric losses resulting from enhanced relaxation polarization. A comparison of the imaginary parts at different frequencies and FeS_2 content yields valuable quantitative insights. In Fig. 6, the imaginary part ϵ'' at 0.8, 1.0, and 1.2 THz were depicted with the variables of FeS_2 content. It is observed that imaginary parts tend to increase with higher frequencies across all coal samples. Furthermore, a robust linear relationship between the imaginary part and FeS_2 content is evident, which can be accurately represented by a linear formula $\epsilon'' = a_2 + b_2x$, as presented in Table 4. The correlation coefficients at 0.8, 1.0, and 1.2 THz are 0.97, 0.97, and 0.98, respectively. These high correlation coefficients establish that FeS_2 content can be reliably estimated using the imaginary part.

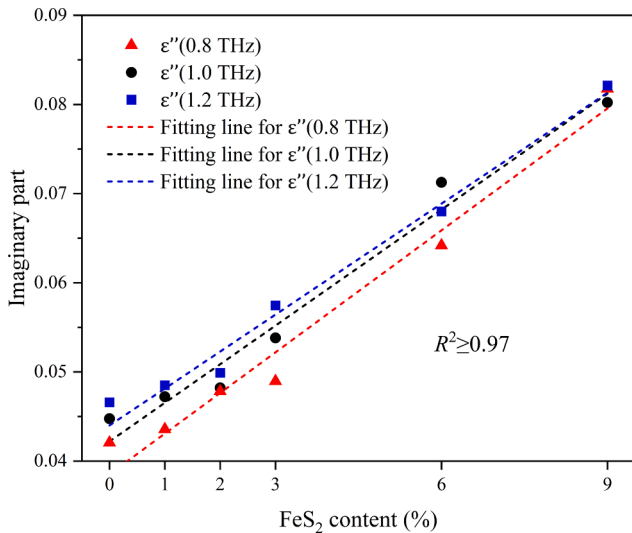


Fig. 6. Variation of Imaginary part values at 0.8, 1.0, and 1.2 THz for lignite versus FeS_2 content.

Table 4

Fitting results of imaginary part and frequency.

Frequency (THz)	$\epsilon'' = a_2 + b_2x$		
	a_2	b_2	R^2
0.8	0.03853	0.00456	0.97
1.0	0.0422	0.00433	0.98
1.2	0.044	0.00415	0.97

3.3. Relationship between terahertz permittivity and characteristic temperature

For lignite samples containing varying FeS_2 content, a notable regularity is observed in both the characteristic temperature and the imaginary part of the dielectric permittivity. Therefore, it becomes feasible to estimate the characteristic temperature using permittivity as a predictor. The relationship between the characteristic temperature and the imaginary part at 1 THz is considered, which can be expressed as follows:

$$T = a_1 + b_1 \frac{\epsilon'' - a_2}{b_2} + c_1 \left(\frac{\epsilon'' - a_2}{b_2} \right)^2 \quad (6)$$

The above formula can be simplified as:

$$T = \left(a_1 + \frac{2c_1a_2 - b_1a_2}{b_2} + \frac{c_1a_2^2}{b_2^2} \right) + \left(\frac{b_1}{b_2} - 2c_1 \right) \epsilon'' + \frac{c_1}{b_2^2} \epsilon''^2 \quad (7)$$

where the coefficients of the first, second, and third terms on the right side of the equation are constant, so the equation can be summarized as follows:

$$T = A + B\epsilon'' + C\epsilon''^2 \quad (8)$$

where A , B , and C are constants, which may be related to the coal type. It is evident that the characteristic temperature is a function represented by a quadratic polynomial of permittivity. Therefore, Fig. 7 illustrates the relationship between characteristic temperature and permittivity, and the experimental data are fitted using Eq. (8). The curve exhibits a robust fitting relationship, and the fitting results are provided in Table 5. Notably, all correlation coefficients (R^2) surpass 0.93, underscoring that the characteristic temperature can be preliminarily estimated by measuring the dielectric permittivity of the coal. It is important to

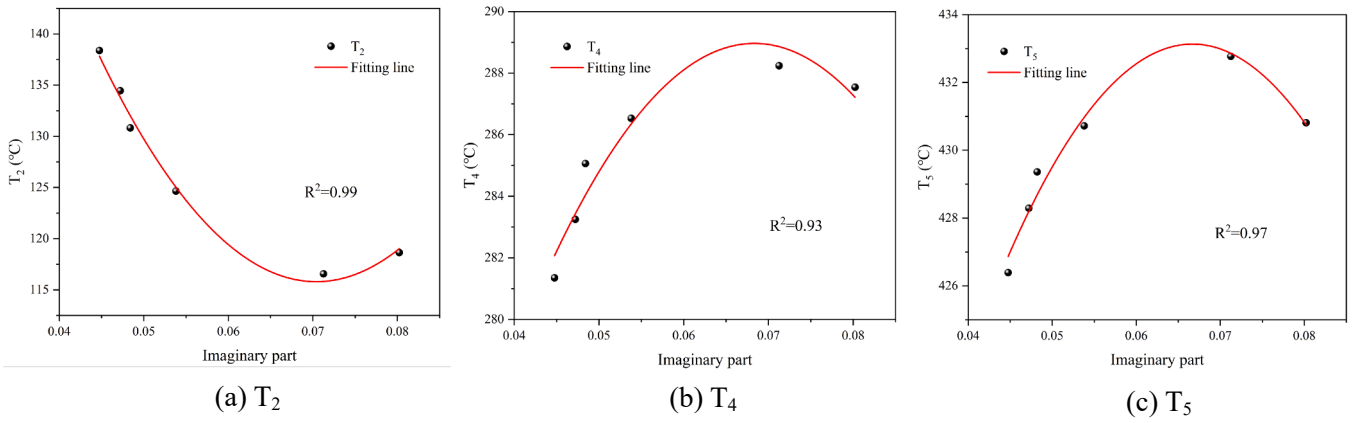


Fig. 7. Variation of characteristic temperature (a) T_2 , (b) T_4 , and (c) T_5 of lignite versus FeS_2 content. Relationship is fitted by function of quadratic polynomial.

Table 5

Fitting results of imaginary permittivity and characteristic temperature.

Characteristic temperature	$T = A + B\varepsilon'' + C\varepsilon''^2$			
	A	B	C	R ²
T_2	281.81	-4715.88	33489.53	0.99
T_4	231.07	1694.15	-12394.63	0.93
T_5	375.25	1735.89	-13014.40	0.97

emphasize that the characteristic temperature maintains a purely numerical relationship with permittivity.

3.4. Verification of relationship and calculation model

(1) Relationship verification based on different rank coals

To verify the quadratic polynomial functional relationship between permittivity and characteristic temperature, TG analysis and THz-TDS analysis were conducted on long flame coal, non-caking coal, and 1/3 coking coal, each with varying pyrite contents (0 %, 3 %, 6 %, and 9 %). The results of the proximate analysis are presented in Table 1. It is noteworthy that non-caking coal and 1/3 coking coal exhibit low moisture content, resulting in inconspicuous dehydration and gas release stages in the TG curves. Therefore, for analysis, T_4 and T_5 were focused. Applying the same data processing method as used for lignite, the relationship between their imaginary permittivity and characteristic temperatures (T_4 and T_5) were established, as depicted in Figs. 8 and 9, respectively. These relationships were also fitted using Eq. (8). The results demonstrate that all correlation coefficients (R^2) exceed 0.85, confirming the accuracy of the quadratic polynomial functional relationship between these two parameters. Furthermore, it is foreseeable

that with an increase in the number of samples, the correlation coefficient values will become even higher.

(2) Calculation model verification based on the lignite

Using the same lignite as an example, FeS_2 powder and pulverized coal were mixed in a random mass ratio to prepare coal samples with randomized FeS_2 content. In total, 36 coal samples were meticulously prepared for subsequent analysis. TG analysis and THz-TDS analysis were conducted on these coal samples with randomized FeS_2 content, yielding their characteristic temperatures and terahertz permittivity. Fig. 10 illustrates the calculated characteristic temperatures using Eq. (8) and the fitting coefficients listed in Table 5, which are then compared to the measured results. Remarkably, the calculated results fall within the ± 1 % error margin of the measured results, with an accuracy of ± 2 °C. This outcome firmly establishes the accuracy of the models utilized for estimating characteristic temperatures based on terahertz permittivity.

China is rich in sulfur-containing coal resources, of which sulfur is mainly in the form of pyrite. Pyrite has a significant effect on the spontaneous combustion of coal. The extremely uneven distribution of pyrite in the coal seam is one of the main reasons for the difference in the spontaneous combustion tendency of coal. However, most studies mainly focused only on the promotion of pyrite on coal spontaneous combustion, ignoring the prerequisite for determining pyrite content. According to the results, the imaginary permittivity has a superior linear relationship with pyrite content, proving the potential of terahertz technology in detecting pyrite. In addition, a numerical quadratic polynomial relationship between the dielectric constant and the characteristic temperature can also be found, because the relationship between pyrite content and the characteristic temperature can be approximated as a quadratic polynomial. This finding is helpful to use

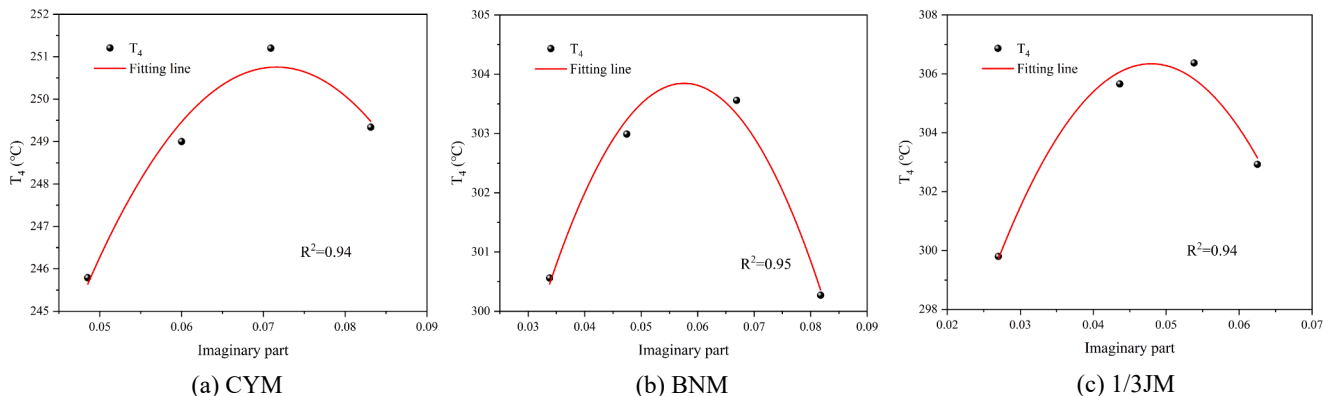


Fig. 8. Variation of characteristic temperature T_4 of (a) CYM, (b) CYM, and (c) 1/3JM versus FeS_2 content. Relationship is fitted by function of quadratic polynomial.

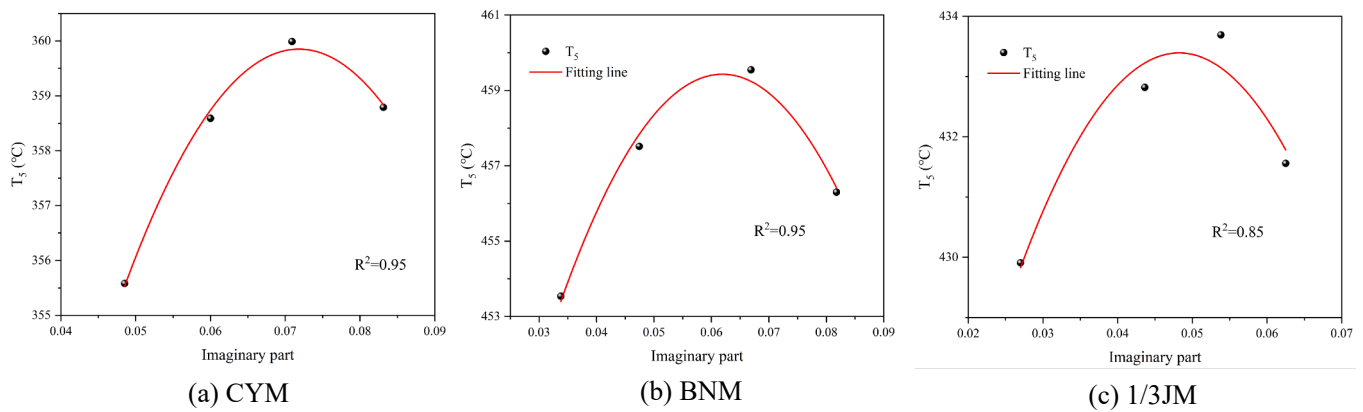


Fig. 9. Variation of characteristic temperature T_5 of (a) CYM, (b) CYM, and (c) 1/3JM versus FeS_2 content. Relationship is fitted by function of quadratic polynomial.

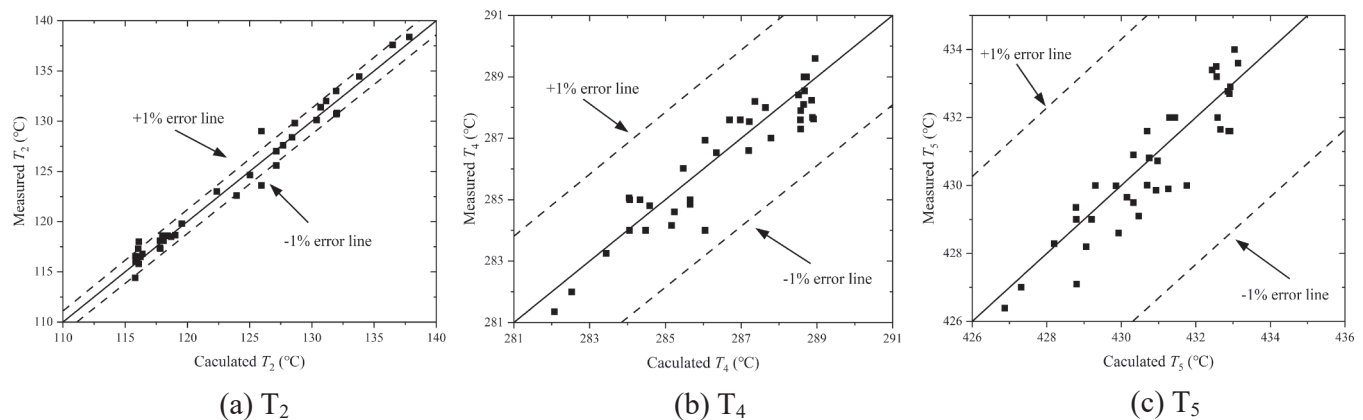


Fig. 10. Comparison of measured results and calculated results: (a) T_2 , (b) T_4 , and (c) T_5 .

terahertz technology to quickly estimate the spontaneous combustion tendency of pyrite-bearing coal, and provides a new idea for controlling the risk of coal spontaneous combustion. However, it should be noted that in many coal mines, coal in different areas of the same seam may have different moisture and ash content, which may affect the permittivity and spontaneous combustion tendency of the coal, which further affects the accuracy of this method for determining the spontaneous combustion tendency. Therefore, to improve the accuracy and applicability, future studies on the effect of moisture and ash on the characteristic temperature and the permittivity should be carried out. In conclusion, although the application of terahertz technology in this area still faces many challenges, there is no doubt that it has potential for monitoring and early warning of spontaneous combustion of coal.

4. Conclusion

In summary, in this study, the characteristic temperatures during the oxidation process of lignite with different pyrite content were determined by a synchronous thermal analyzer. Within the pyrite content range of 0 % to 9 %, pyrite exhibited a pattern of initially enhancing and then diminishing its impact on coal spontaneous combustion. This enhancement effect was manifested in several stages, including the premature termination of the dehydration and degassing stage, the early initiation and delayed termination of the oxygen absorption and weight gain stage, the delayed initiation and premature termination of the oxidative decomposition stage, and the delayed initiation and premature termination of the oxidative combustion stage. This relationship was found to follow a quadratic polynomial function, with a high correlation coefficient ($R^2 > 0.98$). Furthermore, the permittivity of lignite

in the terahertz band was explored using terahertz time-domain spectroscopy. As the pyrite content increased from 0 % to 9 %, the real permittivity exhibited irregular variations, while the imaginary permittivity showed a clear linear increase, supported by a correlation coefficient ($R^2 > 0.97$). This behavior was attributed to the strengthening of relaxation polarization, resulting in more significant dielectric losses. Additionally, a robust quadratic relationship is observed between characteristic temperature and terahertz permittivity. The credibility of this relationship is validated by comparing it with experimental results. These findings contribute to the advancement of the theoretical understanding of coal's permittivity and spontaneous combustion characteristics, and provide a theoretical possibility for the rapid estimation of spontaneous combustion characteristics in pyrite-bearing coal seams using electromagnetic methods.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgments

Funding for this work was provided by the National Natural Science Foundation of China (No.51774290) and PHD Innovative Talent

Cultivation Fund - Basic Research Funds of China University of Mining and Technology-Beijing (No. BBJ202313). The authors would appreciate the anonymous reviewers for their constructive suggestions improving the quality of this paper.

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